Tarea 13

5.43) 24.4L
0.280

$M = 0.400g \Big| \frac{1.25 \times 10^{-3}}{} = 320 g/mol$

5.45) $1 \times 10^{-3} atm \times 1L = n * 0.0821 L * atm * K * 250$

$n = \frac{1 \times 10^{-3} atm * 1L}{0.0821L * 250} = 4.87 \times 10^{-5}$

$4.87 \times 10^{-5} \Big| \frac{6.023 \times 10^{23} mol O_3}{1 mol O_3} = 2.93 \times 10^{19} mol O_3$

5.47) a) $d = \frac{(1atm * l/mol K)(Pm)}{0.0821 * 300.15K} = \frac{(1atm * l/mol K)(0.397 gr/l)}{0.0821 * 300.15°K}$

$= 0.016 g/cm^3 \Big| \frac{1cm^3}{0.001L} = 16 g/L$

b) $n = \frac{(1atm)(2.1L)}{0.0821 * 300.15} = 0.085 mol$ $PM = 0.85 mol * 4.65g = 0.40 mol g$

Escaneado con CamScanner

5.49) $64.9\,g\,C \left| \dfrac{1\,mol}{12.011\,g} \right. = 5.40$ 4

$13.5\,g\,H \left| \dfrac{1\,mol}{1.00794} \right. = 13.61$ 10

$21.6\,g\,O \left| \dfrac{1\,mol}{15.9994} \right. = 1.35$ 1

$FE = C_4H_{10}O$

$m = 74.12\ g/mol$

$M = \dfrac{2.3g * 0.0821 * 393.15}{0.99 * 1}$

$P = 750/760 = 0.99\,atm$

$M = 75\ g/mol$

$74.12/75 = 1$

$FM = C_4H_{10}O$

5.51)

5.53) $2NO + O_2 \rightarrow 2NO_2$

$9L\,NO \left|\frac{1mol\,NO}{30.01L\,NO}\right|\frac{2mol\,NO_2}{2mol\,NO}\left|\frac{30.01L\,NO_2}{1mol\,NO_2}\right. = 9L\,NO_2$

5.55) $S + O_2 \rightarrow SO_2$

$2.54Kg\,S \left|\frac{1000g\,S}{1Kg\,S}\right|\frac{1mol\,SO_2}{32.066g\,S}\left|\frac{22.4L}{1mol\,SO_2}\right|\frac{30.5+273}{273}\left|\frac{1\,L}{1.2atm}\right|\frac{1000mL}{1L} =$

$1,643,810.50mL\,SO_2$

5.57) M del P = 30.97376
F = 18.99840

$2(30.97376) + 4(18.99840) = 137.94$

f. M = P_2F_4

5.59) $NH_3 + HCl \rightarrow NH_4Cl$

$73g\,NH_3 \left|\frac{1mol\,NH_3}{17.03g\,NH_3}\right. = 4.29mol\,NH_3$

$4.29 - 2 = 2.29mol\,NH_3$

$73g\,HCl \left|\frac{1mol\,HCl}{36.46g\,HCl}\right. = 2mol\,HCl$

$$2.29\,g/mol\,NH_3 \times \frac{22.4L}{1mol\,NH_3} \times \frac{287}{273} \times \frac{760}{752} = 54.50\,L\,NH_3$$

5.61) $H_2 + Cl_2 \rightarrow 2HCl$

$$(1)(5.6) = n(0.0821)(273)$$
$$n = 0.25\,mol\,H_2$$

$$0.25 \times \frac{2mol\,HCl}{1mol\,H_2} \times \frac{36.46\,g\,HCl}{1mol\,HC} = 18.23\,g\,HCl$$

5.63)

a) $N_2 + 3H_2 = 2NH_3$

$$12.8L\,NH_3 \times \frac{1mol\,NH_3}{76.02L\,NH_3} \times \frac{1mol\,N_2}{2mol\,NH_3} \times \frac{76.02L\,N_2}{1mol\,N_2} = 6.4L\,N_2$$

$$12.8L\,NH_3 \times \frac{1mol\,NH_3}{76.02L\,NH_3} \times \frac{3mol\,H_2}{2mol\,NH_3} \times \frac{76.02L\,H_2}{1mol\,H_2} = 19.2L\,H_2$$

b) $C_3H_8 + 3H_2O \rightarrow 3CO + 7H_2$

$$8.96L\,H_2 \times \frac{1mol\,H_2}{22.4L\,H_2} \times \frac{1mol\,C_3H_8}{10mol\,H_2} \times \frac{22.4L\,C_3H_8}{1mol\,C_3H_8} = 0.90L\,C_3H_8$$

$$8.96L\,H_2 \times \frac{1mol\,H_2}{22.4L\,H_2} \times \frac{6mol\,H_2O}{10mol\,H_2} \times \frac{22.4L\,H_2O}{1mol\,H_2O} = 53.76L\,H_2O$$